

Name: Sophia Kidwell **Date:** 02-07-2024 **Class:** Block 8

MyDesign Portfolio

Element B Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to structure your research findings around past and current attempts to solve the problem, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own research into prior design attempts, be sure to delete the instructions.

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Pure Enrichment PureZone HEPA Mini Portable Air Purifier>

https://www.bestbuy.com/site/pure-enrichment-true-hepa-portable-air-purifier-blue/6503798.p?skuld=6503798&utm_source=feed



Details

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This is a small portable air filter. It also has a hook on top so it could be hung from a ceiling beam. (Best Buy)

Analysis

One potential downfall is that this air filter is battery powered. Mrs. Poole had stated that she preferred an electric air filter. Other than that, the filter looks good.

Prior Design Attempt #2 - <Corsi-Rosenthal>



[Tex-Air Filters](#)

[5 Filter Corsi-Rosenthal Box MERV 13...](#)

Details

This is a homemade air filter. It is created with 5 air filters attached in a box shape and a box fan as one of the walls of the box. Air is sucked in and all contaminants are caught on the filter before clean air is produced. (Texairfilters.com)

Analysis

This design looks effective at filtering air however it does not meet all requirements. It is way too big for the space, it is not able to be mounted or hung, and it is not portable.

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Prior Design Attempt #3 - <Chemical Insights Air Filter>



<https://www.youtube.com/watch?v=4dLW0sIE6-g>

Details

This is similar to the Corsi-Rosenthal method except it only uses one filter. A filter is attached to the back of a box fan. (Chemical Insights)

Analysis

This is a good design especially because it has an activated carbon filter that absorbs VOC, and Mrs. Poole mentioned this to be a problem. However, it is too big, most likely too loud, and the filter would have to be replaced way too often. It is also too difficult for a student to be able to replace the filter.

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Prior Design Attempt #4 - <bucket air filter>



<https://www.wikihow.com/Make-an-Air-Filter#/Image:Make-an-Air-Filter-Step-16-Version-2.jpg>

Details

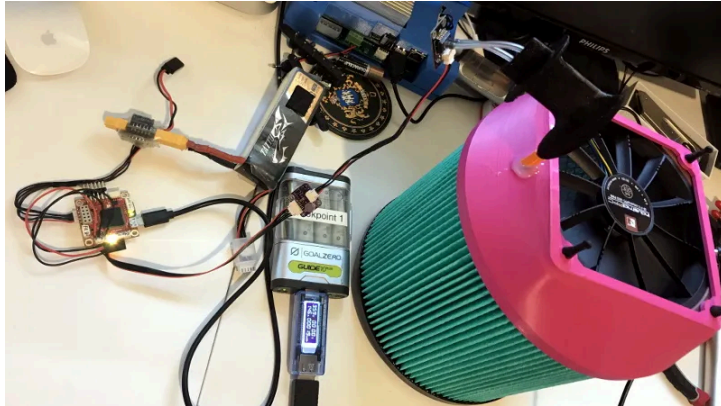
This is made with a bucket and a fan. There are holes cut in the bucket and an air filter put in the bucket. (wiki-how)

Analysis

This would filter air but it would not work for our specifications. For starters, this filter is kind of big. It would also be difficult for a student to change the filter. It would be an obstacle if it was hung from the ceiling and it would not be able to be mounted to the wall. (wiki-how)

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Prior Design Attempt #5 - <DIY Hepa Air Filter>



<https://hackaday.com/2020/01/29/building-and-testing-a-diy-air-purifier/>

Details

This air filter uses a small fan and a round filter to purify air. The air is sucked in through the filter, cleaned, and then sent back out. (Hackaday)

Analysis

This would be a good option for us. It is small, portable, able to be mounted or hung, and seems relatively cheap to make. The only potential problem is the cord not being long enough, but we could use a battery powered fan.